CLASS 9 · JEE MAINS · TOP-10 QUESTION IMPACT REPORT

Pragya G9 (EXT) Achievers

Part Test-1 · which questions cost the top 10 the most

Exam date 08 Oct 2026
Campus HYD/Nallakunta DD Colony
Students analysed top 10 by program rank

In total, the top 10 lost **1,005 marks** – **260** from Maths, **436** from Physics and **309** from Chemistry.



Together they could have scored 2,960 (296 each) and scored 1,955 – the 1,005 marks lost is the gap. Marks lost = marks not earned on wrong or skipped questions plus negative marks.

19 of 74
high-impact questions – lost by 5+ of the top 10

43%
of all lost marks came from Physics

Q27 9/10
most-lost question

196 / 300
top-10 average score (range 166-249)

1 Where the damage happened

Marks the top 10 gave away in each subject, split into marks lost on **wrong answers** (the question's marks plus the negative mark) and marks left on **skipped** questions.

SUBJECT	SPLIT ■ wrong ■ skipped	MARKS LOST	SHARE	ACCURACY*	HIGH-IMPACT
Maths	220 on 44 wrong answers (incl. 44 negative) · 40 on 10 skipped	**260** ≈ 26 per student	26%	82%	**3** of 25 Qs
Physics	300 on 60 wrong answers (incl. 60 negative) · 136 on 34 skipped	**436** ≈ 44 per student	43%	72%	**11** of 25 Qs
Chemistry	245 on 49 wrong answers (incl. 49 negative) · 64 on 16 skipped	**309** ≈ 31 per student	31%	78%	**5** of 24 Qs

* Accuracy = right answers ÷ questions attempted.

BY QUESTION TYPE	MATHS	PHYSICS	CHEMISTRY
Single correct marking +4 / −1	Q1–20 **18%** wrong · **4%** skipped	Q26–45 **26%** wrong · **16%** skipped	Q51–70 **15%** wrong · **5%** skipped
Numerical marking +4 / −1	Q21–25 **16%** wrong · **4%** skipped	Q46–50 **14%** wrong · **6%** skipped	Q71–75 **40%** wrong · **12%** skipped

Key takeaways

1. **Physics** did the most damage: **43%** of all marks lost by the top 10 (436 of 1005). 11 of its 25 questions were lost by 5 or more students.
2. No question beat everyone, but **1** were lost by 9 of 10: Q27.
3. Biggest chapter-level loss: **Chemistry · General Organic Chemistry** – 159 marks across 10 questions (3 high-impact).
4. Skipping is concentrated in **Physics · Single correct**: 31 of 200 answers (16%) were left blank.
5. Negative marking alone cost the top 10 **153 marks** (about 15 per student).
6. Strong ground: all 10 answered **11 questions** correctly (Q2, Q4, Q14, Q16, Q22, Q39, Q44, Q54, Q55, Q56, Q60).

Answered wrong by 5+ students

Students attempted these and still got them wrong – points to a misconception or a trap worth re-teaching.

Q63 8 wrong · **Q27** 7 wrong · **Q31** 7 wrong · **Q71** 7 wrong · **Q7** 6 wrong · **Q20** 6 wrong · **Q28** 6 wrong · **Q43** 6 wrong · **Q45** 6 wrong · **Q66** 6 wrong · **Q35** 5 wrong

Left blank by 5+ students

Most students did not attempt these – points to an unfamiliar topic, low confidence, or running out of time.

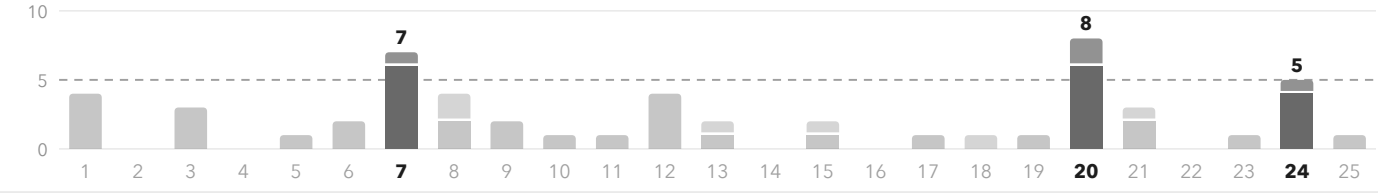
Q26 5 skipped

2 Question-by-question impact

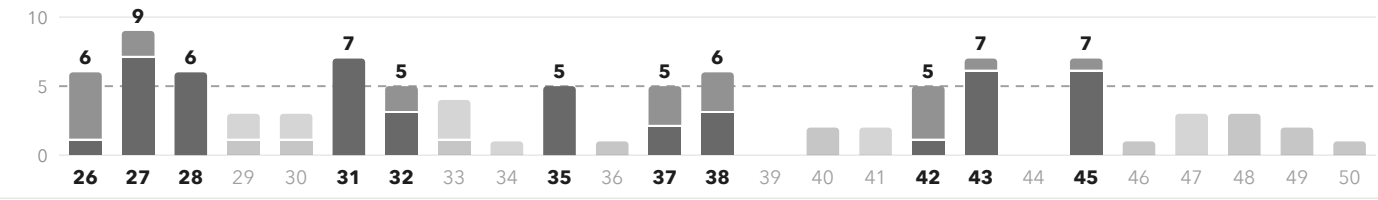
How many of the top 10 lost each question. Questions at or above the dashed line (lost by 5+) are **high-impact** and shown in full colour with their count.

■ Wrong answer ■ Skipped ---- High-impact line (5 of 10) ■ Below the line (faded) "del" = deleted / bonus question

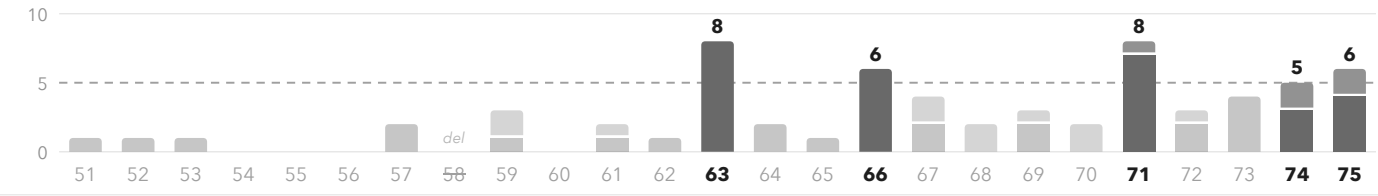
Maths — **3** of 25 questions high-impact · **260** marks lost by the top 10



Physics — **11** of 25 questions high-impact · **436** marks lost by the top 10



Chemistry — **5** of 24 questions high-impact · **309** marks lost by the top 10



3 Chapter hot-spots

Marks lost by the top 10, chapter by chapter. Subjects are ordered from most marks lost to least; within each subject, re-teach from the top.

Marks lost adds up what all 10 students gave away on a chapter's questions: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. Example: **Chemistry · General Organic Chemistry** has 10 questions, and the top 10 lost **159** marks on them — about **16 per student**. **Students per Q** = on average, how many of the 10 lost each question in that chapter (here 3.4 of 10).

CHAPTER	QS	HIGH-IMPACT QUESTIONS	MARKS LOST BY TOP 10	PER STUDENT	STUDENTS PER Q
1. Physics 436 marks lost · 43% of all lost marks · 11 high-impact questions					
Work, Energy & Power	5	**Q31, Q38, Q42, Q45**	**121**	≈ 12 marks	5.2 of 10
Laws of Motion	8	**Q37, Q43**	**118**	≈ 12 marks	3.2 of 10
Circular Motion	7	**Q27, Q32, Q35**	**109**	≈ 11 marks	3.3 of 10
Momentum & Collisions	3	**Q26**	**45**	≈ 4 marks	3.3 of 10
Friction	2	**Q28**	**43**	≈ 4 marks	4.5 of 10
2. Chemistry 309 marks lost · 31% of all lost marks · 5 high-impact questions					
General Organic Chemistry	10	**Q66, Q74, Q75**	**159**	≈ 16 marks	3.4 of 10
Isomerism	10	**Q71**	**77**	≈ 8 marks	1.6 of 10
Hydrocarbons	4	**Q63**	**73**	≈ 7 marks	3.8 of 10
3. Maths 260 marks lost · 26% of all lost marks · 3 high-impact questions					
Functions	7	**Q24**	**96**	≈ 10 marks	2.9 of 10
Inverse Trigonometric Functions	10	**Q20**	**93**	≈ 9 marks	1.9 of 10
Trigonometric Equations	7	**Q7**	**67**	≈ 7 marks	2.0 of 10
Sequences & Series	1	None reached 5 · most missed Q18 (1/10)	**4**	≈ 0 marks	1.0 of 10

4 High-impact questions at a glance

Every question lost by 5 or more of the top 10, one table per subject, worst first. The cropped question for each is in section 7.

Marks lost = marks the top 10 together gave away on that question: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. Example: **Q27** – 7 wrong × 5 + 2 skipped × 4 = 43 marks. **Lost by** = how many of the 10 got it wrong or skipped it.

Critical lost by 9–10 **High** lost by 7–8 **Moderate** lost by 5–6

Maths

3 of 25 questions high-impact · **96** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q20	**Inverse Trigonometric Functions** – Composite inverse-trig function – counting solutions	Single · +4 / −1	8/10	6	2	38
Q7	**Trigonometric Equations** – $\sqrt{3}\sin nx + \cos nx$ = quadratic – number of solutions	Single · +4 / −1	7/10	6	1	34
Q24	**Functions** – Even function with GIF, fractional part and signum	Numerical · +4 / −1	5/10	4	1	24

Physics

11 of 25 questions high-impact · **319** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q27	**Circular Motion** – Coins on a rotating disc – friction limit at each radius	Single · +4 / −1	9/10	7	2	43
Q31	**Work, Energy & Power** – KE vs height for a stone thrown up from a cliff	Single · +4 / −1	7/10	7	0	35
Q43	**Laws of Motion** – Tension in a spring pulled by 5 N at each end	Single · +4 / −1	7/10	6	1	34
Q45	**Work, Energy & Power** – Speed on hitting the ground – energy conservation	Single · +4 / −1	7/10	6	1	34
Q28	**Friction** – Friction vs incline angle graph (angle of repose)	Single · +4 / −1	6/10	6	0	30
Q38	**Work, Energy & Power** – Work along different paths; conservative or not	Single · +4 / −1	6/10	3	3	27
Q26	**Momentum & Collisions** – KE loss in head-on collision from relative velocities	Single · +4 / −1	6/10	1	5	25
Q35	**Circular Motion** – Power delivered by the centripetal force	Single · +4 / −1	5/10	5	0	25
Q32	**Circular Motion** – Blocks on a rotating frame – ω when sliding starts	Single · +4 / −1	5/10	3	2	23
Q37	**Laws of Motion** – Tension in a pulley–block system	Single · +4 / −1	5/10	2	3	22
Q42	**Work, Energy & Power** – Change in PE of a ball thrown from a tower	Single · +4 / −1	5/10	1	4	21

Chemistry

5 of 24 questions high-impact · **160** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q63	**Hydrocarbons** – Monochlorination products incl. stereoisomers	Single · +4 / −1	8/10	8	0	40
Q71	**Isomerism** – Count of optically active compounds	Numerical · +4 / −1	8/10	7	1	39
Q66	**General Organic Chemistry** – False order of −I effect	Single · +4 / −1	6/10	6	0	30
Q75	**General Organic Chemistry** – Bases stronger than aniline	Numerical · +4 / −1	6/10	4	2	28
Q74	**General Organic Chemistry** – Compounds giving a positive bicarbonate test	Numerical · +4 / −1	5/10	3	2	23

5 Who lost which high-impact question

Each row is a student (campus order); each column a high-impact question. **Prog. rank** = rank in the program, **AIR** = All-India rank, **Total** = marks out of 300, **Avg** = average of the 6 weekly Mains tests (WTM), out of 300. ✗ = wrong, – = skipped, ✓ = correct.

■ ✗ Wrong ■ – Skipped ■ ✓ Correct

	Rank & score				Maths			Physics										Chemistry					
STUDENT	PROG. RANK	AIR	TOTAL	AVG /300	7	20	24	26	27	28	31	32	35	37	38	42	43	45	63	66	71	74	75	Lost
Geetansh P. R. Vengala	1	**3**	249	196.8*	✓	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗	✓	✓	**6**
Puneeth K. Jureddi	2	**20**	220	227.3*	✓	✗	✓	–	✗	✗	✓	✓	✗	✓	–	–	✗	–	✗	✓	✗	✓	✓	**11**
Nandimandalam L. Reddy	3	**22**	216	191.8*	–	–	✓	✓	✗	✗	✓	–	✓	✓	–	–	✗	✓	✗	✗	✗	✓	✗	**12**
Masarapu Kartikeya	4	**31**	207	194.8	✗	✓	✗	–	✓	✓	✗	✗	✓	✓	–	–	–	✗	✗	✓	–	✓	–	**12**
Ramini Havisha	6	**48**	196	199.8	✗	–	–	–	–	✗	✗	–	✓	–	✓	✓	✓	✗	✗	✗	✓	–	–	**14**
Prit Raj Kushwaha	9	**58**	186	200.0*	✓	✓	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓	✗	**8**
Dharavath Joseph	10	**76**	179	179.9*	✗	✗	✓	–	✗	✗	✗	✓	✗	✗	✓	–	✗	✗	✗	✓	✗	–	✗	**15**
Pulipati Vikas	14	**107**	170	160.8*	✗	✗	✗	✓	✗	✓	✗	✗	✗	–	✗	✓	✗	✗	✓	✗	✗	✗	✓	**14**
Rachoori Vishwateja	16	**122**	166	183.7*	✗	✗	✓	–	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	**16**
Vyasabhattu Shreyasraju	17	**123**	166	162.9	✗	✗	✗	✓	–	✓	✗	✗	✗	–	✗	✓	✗	✓	✗	✗	✓	✗	✓	**13**
Students who lost it					7	8	5	6	9	6	7	5	5	5	6	5	7	7	8	6	8	5	6	

6 Subject scores of the top 10

Scores rebuilt from each student's right / wrong / skipped answers (they match the official totals exactly). The weakest subject (lowest % of its maximum) is highlighted. **Subject maximum** leaves out the deleted / ungraded questions: Chemistry 96 = 100 minus Q58 (4).

STUDENT	PROG. RANK	AIR	TOTAL	AVG /300	Maths score · right / wrong / skipped	Physics score · right / wrong / skipped	Chemistry score · right / wrong / skipped	WEAKEST
Geetansh P. R. Vengala	1	**3**	**249**	196.8*	**95**/100 24 / 1 / 0	**90**/100 23 / 2 / 0	**64**/96 17 / 4 / 3	Chemistry
Puneeth K. Jureddi	2	**20**	**220**	227.3*	**90**/100 23 / 2 / 0	**49**/100 14 / 7 / 4	**81**/96 21 / 3 / 0	Physics
Nandimandalam L. Reddy	3	**22**	**216**	191.8*	**88**/100 22 / 0 / 3	**65**/100 17 / 3 / 5	**63**/96 17 / 5 / 2	Physics
Masarapu Kartikeya	4	**31**	**207**	194.8	**80**/100 21 / 4 / 0	**52**/100 14 / 4 / 7	**75**/96 19 / 1 / 4	Physics
Ramini Havisha	6	**48**	**196**	199.8	**69**/100 18 / 3 / 4	**49**/100 13 / 3 / 9	**78**/96 20 / 2 / 2	Physics
Prit Raj Kushwaha	9	**58**	**186**	200.0*	**65**/100 18 / 7 / 0	**70**/100 19 / 6 / 0	**51**/96 15 / 9 / 0	Chemistry
Dharavath Joseph	10	**76**	**179**	179.9*	**70**/100 19 / 6 / 0	**47**/100 14 / 9 / 2	**62**/96 17 / 6 / 1	Physics
Pulipati Vikas	14	**107**	**170**	160.8*	**61**/100 17 / 7 / 1	**52**/100 15 / 8 / 2	**57**/96 16 / 7 / 1	Physics
Rachoori Vishwateja	16	**122**	**166**	183.7*	**62**/100 17 / 6 / 2	**37**/100 12 / 11 / 2	**67**/96 18 / 5 / 1	Physics
Vyasabhattu Shreyasraju	17	**123**	**166**	162.9	**60**/100 17 / 8 / 0	**53**/100 15 / 7 / 3	**53**/96 15 / 7 / 2	Physics

Average. "Avg" = the student's average over the 6 weekly Mains tests (WTM) before this Part Test, out of 300, from the campus Mains & Advance averages sheet. * Sat fewer weekly tests (tests sat / conducted): Geetansh Praveen Reddy Vengala (5/6), Puneeth Karthikeya Jureddi (5/6), Nandimandalam Loukhya Reddy (5/6), Prit Raj Kushwaha (5/6), Dharavath Joseph (5/6), Pulipati Vikas (5/6), Rachoori Vishwateja (5/6).

Ranks. Program rank = rank within Pragya G9 (EXT) Achievers across all campuses (from the final marks sheet; the top 10 are picked by this rank). **AIR** = All-India rank across the whole Class 9 programme (all batches, all campuses), from the institute's All India Marks Analysis sheet.

How to read this report. "Lost" means the student did not get the question right – either answered it wrong or left it blank. A question is **high-impact** when 5 or more of the top 10 lost it. **"Marks lost"** = marks the top 10 together gave away: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. "Per student" divides that by 10. **Deleted question:** Q58 has no right / wrong / skipped entry for any student in the final marks sheet (deleted / bonus question), so it is left out of all counts and shown as "del" in the charts. Chapter tags were assigned by reading each question; please correct any that your department classifies differently.

7 Maths – high-impact questions

3 questions, worst first
cropped from the question paper

Q20 Inverse Trigonometric Functions – Composite inverse-trig function – counting solutions

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 + 2 skipped × 4 = 38

Lost by 8/10
6 wrong · 2 skipped

20. Consider the functions $rage(x)=\sin^{-1}\sqrt{x}+e^{\left(\tan^{-1}\left(\sqrt{x-1}\right)\right)^2}+\left(\cos^{-1}\left(\sqrt{x}\right)\right)^3$ and $bait(x)=x^2+x+1$ within their respective domains. Then the number of solutions of $rage(bait(x))<3$ is/are ……..

A) 0 B)1 C)2 D)3

Wrong: Geetansh Praveen Reddy Vengala, Puneeth Karthikeya Jureddi, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Skipped: Nandimandalam Loukhya Reddy, Ramini Havisha

Q7 Trigonometric Equations – √3 sin πx + cos πx = quadratic – number of solutions

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 + 1 skipped × 4 = 34

Lost by 7/10
6 wrong · 1 skipped

7. Number of solutions of $\sqrt{3}\sin(\pi x)+\cos(\pi x)=x^2-\frac{2}{3}x+\frac{19}{9}$ is ……..

A) 0 B) 1 C) 2 D) 3

Wrong: Masarapu Kartikeya, Ramini Havisha, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Skipped: Nandimandalam Loukhya Reddy

Q24 Functions – Even function with GIF, fractional part and signum

Maths · Numerical · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 1 skipped × 4 = 24

Lost by 5/10
4 wrong · 1 skipped

24. Let $f(x)=\left([a]^2-5[a]+4\right)x^3-\left(6\{a\}^2-5\{a\}+1\right)x-(\tan x)(\operatorname{sgn}(x))$ (where [.] = GIF, $\{.\}=F.P.F, \operatorname{sgn}(x)=signum(x)$). If $f(x)$ is an even function of 'x', then the sum of possible values of 'a' is ……?

Wrong: Masarapu Kartikeya, Prit Raj Kushwaha, Pulipati Vikas, Vyasabhattu Shreyasraju

Skipped: Ramini Havisha

Physics – high-impact questions

11 questions, worst first
cropped from the question paper

Q27 Circular Motion – Coins on a rotating disc – friction limit at each radius

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 7 wrong × 5 + 2 skipped × 4 = 43

Lost by 9/10
7 wrong · 2 skipped

27. A disc revolves with a speed of 100/3 rev/min and has radius of 15cm. Two coins of same mass as disc are placed at 4 cm and 14cm away from the centre of disc. If the coefficient of friction between the coins and the disc is 0.15, which of the two coins will revolve with disc?
A) both the coins will revolve
B) No coin will revolve with disc
C) coin at 4cm will revolve with disc
D) coin at 14cm will revolve with disc

Wrong: Geetansh Praveen Reddy Vengala, Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Prit Raj Kushwaha, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja

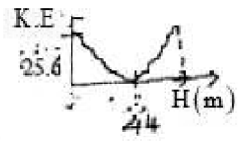
Skipped: Ramini Havisha, Vyasabhattu Shreyasraju

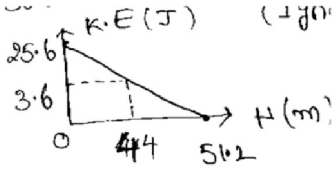
Q31 Work, Energy & Power – KE vs height for a stone thrown up from a cliff

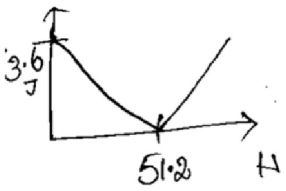
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 7 wrong × 5 = 35

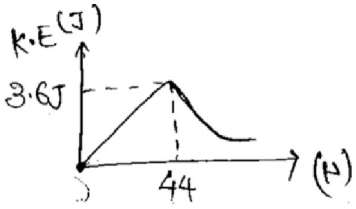
Lost by 7/10
7 wrong · 0 skipped

31. A stone of mass 50g is thrown vertically upwards with a velocity of 12m/s from edge of a cliff 44m high. Identify kinetic energy of stone versus height from bottom of cliff $\left(g = 10 m/s^2\right)$(ignore air resistance)

A)


B)


C)


D)


Wrong: Masarapu Kartikeya, Ramini Havisha, Prit Raj Kushwaha, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

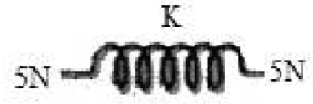
Skipped: –

Q43 Laws of Motion – Tension in a spring pulled by 5 N at each end

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 + 1 skipped × 4 = 34

Lost by 7/10
6 wrong · 1 skipped

43. The tension in the massless spring is



A) zero B) 10N C) 2.5N D) 5N

Wrong: Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Skipped: Masarapu Kartikeya

Q45 **Work, Energy & Power** – Speed on hitting the ground – energy conservation
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 + 1 skipped × 4 = 34

Lost by 7/10
6 wrong · 1 skipped

45. A cricket bowler releases the ball in two different ways

i) giving it only horizontal velocity and

ii) giving it horizontal velocity and small downward velocity

The speeds at the time release is same. Both are released at a height H from ground then which one will have greater speed when ball hits the ground?(Neglect air resistance)

A) Ist way B) IInd way C) same in two ways D) None of the above

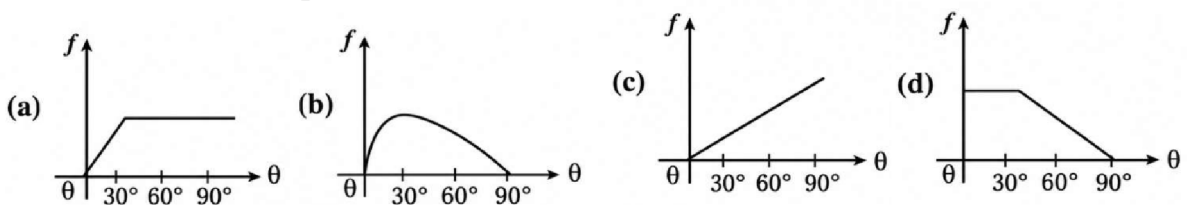
Wrong: Geetansh Praveen Reddy Vengala, Masarapu Kartikeya, Ramini Havisha, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja

Skipped: Puneeth Karthikeya Jureddi

Q28 **Friction** – Friction vs incline angle graph (angle of repose)
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 = 30

Lost by 6/10
6 wrong · 0 skipped

28. A block is placed on a rough plane whose inclination to the horizontal (θ) can be varied. The angle of repose is 30°. The graph that correctly indicate the variation of frictional force (f) between block and the plane with θ is



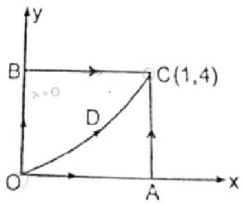
Wrong: Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Ramini Havisha, Prit Raj Kushwaha, Dharavath Joseph, Rachoori Vishwateja

Skipped: –

Q38 **Work, Energy & Power** – Work along different paths; conservative or not
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 3 wrong × 5 + 3 skipped × 4 = 27

Lost by 6/10
3 wrong · 3 skipped

38. A particle is moved along the different paths OAC, OBC and ODC as shown in the fig. path ODC is parabola, $y = 4x^2$. The work done by a force $\vec{F} = xy\hat{i} + x^2 y\,\hat{j}$ on the particle along



Column – I	Column – II
a)Path OAC	P) 2J
b) Path OBC	Q) $\frac{19}{3}$J
c) Path ODC	R) Conservative
d) Force is	S) 8J
	T) Non- conservative

A) $a \to Q, b \to S, c \to P, d \to T$ B) $a \to S, b \to P, c \to Q, d \to T$

C) $a \to S, b \to Q, c \to P, d \to T$ D) $a \to S, b \to Q, c \to S, d \to T$

Wrong: Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Skipped: Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Masarapu Kartikeya

Q26 **Momentum & Collisions** – KE loss in head-on collision from relative velocities
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 1 wrong × 5 + 5 skipped × 4 = 25

Lost by 6/10
1 wrong · 5 skipped

26. Two bodies of masses 3 kg and 2 kg collide head on. Their relative velocities before and after collision are 15m/s and 5 m/s respectively. The loss of kinetic energy of system is
A) 120J B) 100J C) 80J D) 240J

Wrong: Prit Raj Kushwaha

Skipped: Puneeth Karthikeya Jureddi, Masarapu Kartikeya, Ramini Havisha, Dharavath Joseph, Rachoori Vishwateja

Q35 **Circular Motion** – Power delivered by the centripetal force
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 5 wrong × 5 = 25

Lost by 5/10
5 wrong · 0 skipped

35. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration changing with time 't' as $a_c = k^2 t^2$, where K is a constant. The power delivered to the particle by centripetal force acting on it is

A) zero B) $mk^2 rt$ C) $mk^2 r^2 t$ D) $mk^2 \sqrt{rt}$

Wrong: Puneeth Karthikeya Jureddi, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Skipped: –

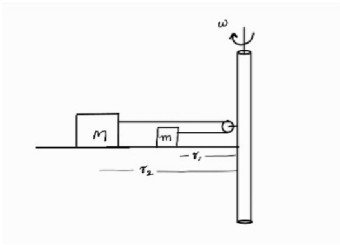
Q32 Circular Motion – Blocks on a rotating frame – ω when sliding starts

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 3 wrong × 5 + 2 skipped × 4 = 23

Lost by 5/10

3 wrong · 2 skipped

32. Two blocks of masses m and M are connected by a chord passing around a frictionless pulley which is attached to a rotating frame, which rotates about a vertical axis with angular velocity ω. If co-efficient of friction between two masses and the surface be μ_1 and μ_2 respectively. The value of ω at which block starts sliding radially $(M > m)$



A) $\omega = \left[g \left[\frac{\mu_1 M + \mu_2 m}{m r_1 - M r_1} \right] \right]^{1/2}$

B) $\omega = \left[g \left[\frac{\mu_1 m + \mu_2 M}{M r_2 - m r_1} \right] \right]^{1/2}$

C) $\omega = \left[g \left[\frac{\mu_1 m - \mu_2 M}{M r_2 - m r_1} \right] \right]^{1/2}$

D) $\omega = \left[g \left[\frac{\mu_1 M - \mu_2 m}{M r_2 - m r_1} \right] \right]^{1/2}$

Wrong: Masarapu Kartikeya, Pulipati Vikas, Vyasabhattu Shreyasraju

Skipped: Nandimandalam Loukhya Reddy, Ramini Havisha

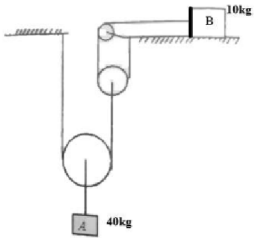
Q37 Laws of Motion – Tension in a pulley-block system

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 2 wrong × 5 + 3 skipped × 4 = 22

Lost by 5/10

2 wrong · 3 skipped

37. Fig shows two blocks A and B connected to an ideal pulley string system. In this system when bodies are released then $\left(g = 10 m/s^2\right)$(Mass of a is 40Kg and B is 10Kg on smooth surface). The tension in the string connecting the block B is



A) $20N$ B) $80N$ C) $60N$ D) $40N$

Wrong: Dharavath Joseph, Rachoori Vishwateja

Skipped: Ramini Havisha, Pulipati Vikas, Vyasabhattu Shreyasraju

Q42 **Work, Energy & Power** – **Change in PE of a ball thrown from a tower**
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 1 wrong × 5 + 4 skipped × 4 = 21

Lost by 5/10
1 wrong · 4 skipped

42. At t =0 sec a 1kg ball is thrown from top to a tall tower with velocity $\vec{V} = \left(8\hat{i} + 9\hat{j}\right)$ m/s. The change in potential energy of ball – earth system between $t = 0$ *to* $t = 5\text{sec}$ is

$\left(g = 10m/s^2\right)$

A) 800J B) 1536J C) 2100J D) -800J

Wrong: Rachoori Vishwateja

Skipped: Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Masarapu Kartikeya, Dharavath Joseph

Chemistry – high-impact questions

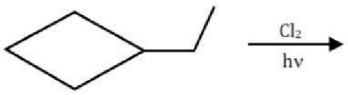
5 questions, worst first
cropped from the question paper

Q63 Hydrocarbons – Monochlorination products incl. stereoisomers

Chemistry · Single correct · marking +4 / −1 · marks lost by the top 10: 8 wrong × 5 = 40

Lost by 8/10
8 wrong · 0 skipped

63. Number of monochlorinated product when following compound undergo reaction with $Cl_2 / h\nu$ (including stereo isomers)



A) 5 B) 9 C) 10 D) 6

Wrong: Geetansh Praveen Reddy Vengala, Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Masarapu Kartikeya, Ramini Havisha, Dharavath Joseph, Rachoori Vishwateja, Vyasabhattu Shreyasraju

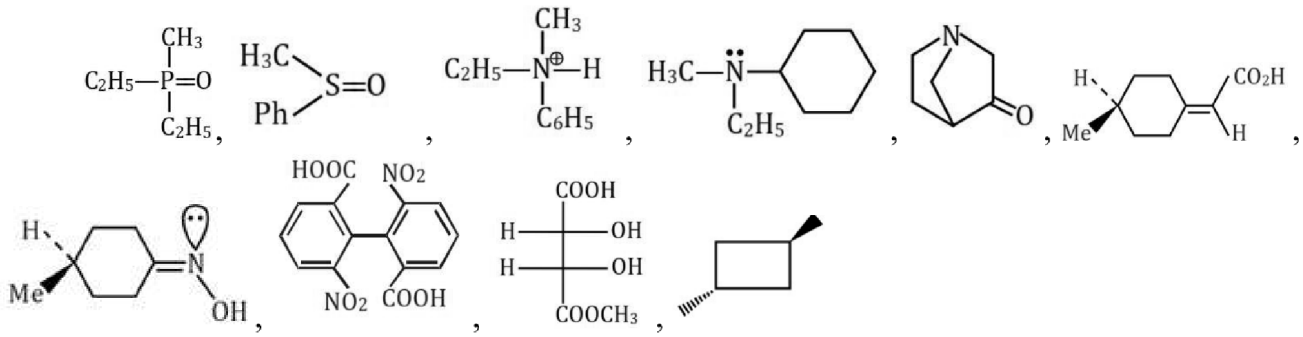
Skipped: –

Q71 Isomerism – Count of optically active compounds

Chemistry · Numerical · marking +4 / −1 · marks lost by the top 10: 7 wrong × 5 + 1 skipped × 4 = 39

Lost by 8/10
7 wrong · 1 skipped

71. Total no of compounds which are optically active are _____



Wrong: Geetansh Praveen Reddy Vengala, Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Prit Raj Kushwaha, Dharavath Joseph, Pulipati Vikas, Rachoori Vishwateja

Skipped: Masarapu Kartikeya

Q66 General Organic Chemistry – False order of –I effect

Chemistry · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 = 30

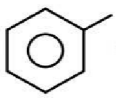
Lost by 6/10
6 wrong · 0 skipped

66. Which of the following is false order of – I effect ?

A) $-F > -Cl > -Br > -I$

B) $-\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2$

C) $-F > -OH > -NH_2$

D)  $> -C \equiv CH > -H$

Wrong: Geetansh Praveen Reddy Vengala, Nandimandalam Loukhya Reddy, Ramini Havisha, Prit Raj Kushwaha, Pulipati Vikas, Vyasabhattu Shreyasraju

Skipped: –

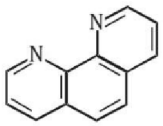
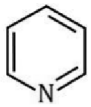
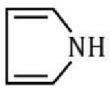
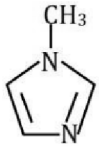
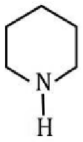
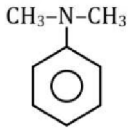
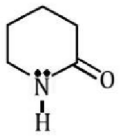
Q75 General Organic Chemistry – Bases stronger than aniline

Chemistry · Numerical · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 2 skipped × 4 = 28

Lost by 6/10

■ 4 wrong ■ 2 skipped

75. How many of the following are more basic then Aniline?

NH_3, $PhCH_2NH_2$,  ,  ,  ,  ,  ,  , $CH_3—CH_2—NH_2$, $CH_3—C\equiv N$, 

Wrong: Nandimandalam Loukhya Reddy, Prit Raj Kushwaha, Dharavath Joseph, Rachoori Vishwateja

Skipped: Masarapu Kartikeya, Ramini Havisha

Q74 General Organic Chemistry – Compounds giving a positive bicarbonate test

Chemistry · Numerical · marking +4 / −1 · marks lost by the top 10: 3 wrong × 5 + 2 skipped × 4 = 23

Lost by 5/10

■ 3 wrong ■ 2 skipped

74. How many of the following compounds can give positive bicarbonate test?

HCl, Picric Acid, Benzoic acid, 2,4-dintrophenol, 4-nitrophenol, Squaric Acid, formic Acid, Benzene Sulphuric acid, 1-Butyne, NH_4Cl .

Wrong: Pulipati Vikas, Rachoori Vishwateja, Vyasabhattu Shreyasraju

Skipped: Ramini Havisha, Dharavath Joseph